Structure formation in an expanding universe with a numerical-relativity particle-in-cell code: Working paper

Author list to be set
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This working paper records the extension of a numerical-relativity particle-in-cell code to cosmological structure formation. With that code it redoes the comparison of general-relativistic and Newtonian collapse of East, Wojtak, and Pretorius, the integration tests of Daverio, Dirian, and Mitsou, and the initial-condition tests of Barrera-Hinojosa and Li. The code reproduces the homogeneous matter-dominated universe, the linear growth of a horizon-scale perturbation, and the luminosity distance along light rays. In the collapse of a perturbation at one wavelength, run with identical particles and resolution in the two theories of gravity, the relativistic infall is later by a fraction of about three times the depth of the initial potential. This horizon-scale effect is converged with resolution and is smaller than the delay reported by East, Wojtak, and Pretorius for the lower amplitudes, whose two codes differed in resolution. Light rays through the collapsing universe reproduce the luminosity distances of that work. The integration tests of Daverio, Dirian, and Mitsou over a thousandfold expansion are passed for the homogeneous universe, for a linear mode, and for a random field drawn from a power spectrum, and the initial-condition tests of Barrera-Hinojosa and Li are reproduced without radiation. Five findings concern the method. Matter sources held fixed during a spacetime step are a first-order error in an expanding universe. The deposit of a cold particle lattice carries an error set by its compression and by the order of the particle cloud. The cloud weakens the self-gravity of short waves, and a compensation pass on the deposited density, applied before the division by the volume element, restores their growth. The cubic Hermite reconstruction of the metric at the particles has a gradient error of second order that depends on the position of a particle within its cell; the cubic spline through the cell values removes it, and the energy and momentum of the deposited matter are then conserved on the mesh at fourth order. The error of the time integrator on the expanding background is committed in the first steps and stays as a uniform constraint violation unless the step is limited by the expansion rate. The progress ledger in the appendix states what is done and what is open.

I. AIM

Cold dark matter is collisionless: after the first crossing of matter streams it has many velocities at one place, which a fluid cannot represent. A particle description coupled to the full Einstein equations follows it through the formation of bound structures, and answers how much a Newtonian N-body calculation on a prescribed expanding background misses. East, Wojtak, and Pretorius [1], building on Ref. [2], made that comparison for inhomogeneities at one wavelength with amplitudes from the cosmological to the extreme. Daverio, Dirian, and Mitsou [3] tested the time integration of a relativistic N-body code, and Barrera-Hinojosa and Li [4, 5] built and tested initial conditions for one.

The question of this work is whether the numerical-relativity particle-in-cell method of Ref. [6], as rebuilt in our earlier work for isolated systems, reproduces these results when it is turned to an expanding, periodic universe, and what the method needs for that. Section II writes the expanding universe in the variables of the code and shows why the calculation is inexpensive in conformal time. Section III gives the additions to the method. Section IV holds the results. Section V collects what the tests say about the method.

II. AN EXPANDING UNIVERSE IN THE VARIABLES OF THE CODE

The code evolves the spatial metric γ_{ij} and the extrinsic curvature K_{ij} of the slices of constant time t, with lapse α and shift β^i, in the Z4c formulation, and moves macroparticles of rest mass m_p along geodesics [6]. A flat, matter-dominated homogeneous universe with scale factor a has

$$\gamma_{ij} = a^2 \delta_{ij}, \qquad K = -\frac{3}{\alpha}\frac{\dot{a}}{a} = -3H, \qquad H^2 = \frac{8\pi}{3}\rho, \tag{1}$$

where K is the trace of K_{ij}, a dot is a derivative with respect to t, H is the Hubble rate, and ρ is the density. Matter at rest in these comoving coordinates stays at rest, and its density falls as a^{-3}.

The slicing decides the cost. Gravitational fields travel at the speed of light, so a time step may cover at most a fraction of a cell. With the lapse condition

$$\partial_t \alpha = -\tfrac{1}{3}\,\alpha^2 K, \tag{2}$$

a member of the Bona–Massó family, the homogeneous solution has $\alpha = a$. The time coordinate is then conformal time, light covers one comoving cell per unit of $t/\Delta x$ at all epochs, and a fixed time step remains stable while the universe expands. In a matter-dominated universe $a \propto t^2$, so an expansion by a factor a takes $2(\sqrt{a}-1)/H_0$ of conformal time, where H_0 is the initial Hubble rate.

For a periodic box of four initial Hubble radii, the choice of Ref. [1], the expansion to the collapse of a one-percent perturbation is a few box-crossing times, hence about a thousand time steps on a mesh of a hundred cells per side. The shift is zero. The mesh is uniform and periodic, and the spatial coordinates are comoving.

III. METHOD

A. Periodic domain and particles as observers

Particles that leave the box re-enter on the opposite side, and the clouds that deposit their energy, momentum, and stress on the mesh wrap around in the same way. Every particle carries its proper time τ, advanced with the lapse and Lorentz factor at the midpoint of each step, so that any particle serves as a fiducial observer. Particles of zero rest mass follow null geodesics with the same pusher and carry no sources; they serve as light rays. A ray that is emitted during the run waits at its point of emission, and at the emission time its direction becomes the null momentum of unit frequency for the normal observer there.

B. Matter sources in an expanding universe

The coupling of Ref. [6] deposits the energy density E, the momentum density S_i, and the stress S_{ij} seen by normal observers once per time step and holds them fixed during the stages of the spacetime integrator. Each is a sum over particles divided by the volume element $\sqrt{\gamma}$ of the cell. Around an isolated object the volume element changes little during a step. In an expanding universe it grows at three times the Hubble rate, and sources held fixed are a first-order error in time with that coefficient. The particle sums change only through the motion and the Lorentz factors of the particles, which is slow. The code therefore hands the spacetime solver the densities

$$\hat{E} = \sqrt{\gamma}\, E, \quad \hat{S}_i = \sqrt{\gamma}\, S_i, \quad \hat{S}_{ij} = \sqrt{\gamma}\, S_{ij}, \tag{3}$$

and the solver divides them by the volume element of each stage, $1/\sqrt{\gamma} = \chi^{3/2}$ with the conformal factor χ of the Z4c variables. The densities are extrapolated to the middle of the step from the last two deposits,

$$\hat{E}^{\,n+1/2} = \hat{E}^{\,n} + \frac{\Delta t_{n+1}}{2\,\Delta t_n}\left(\hat{E}^{\,n} - \hat{E}^{\,n-1}\right), \tag{4}$$

where Δt_n is the step that ends at the deposit n, and likewise for the other two, which makes the coupling second order for the slow part as well. Daverio, Dirian, and Mitsou refresh the same factor between particle updates [3].

The cloud that spreads a particle over the mesh smooths the deposited densities. A wave of wave number k on a mesh of spacing Δx is lowered by the factor $1 - \frac{1}{2}s^2(k\Delta x)^2$, where $s^2\Delta x^2$ is the second moment of the cloud in each direction: $s^2 = 1/6$, $1/4$, and $1/3$ for the linear, quadratic, and cubic cloud. The smoothing weakens the self-gravity of the wave and slows its growth. One pass over the deposited densities with the weights $(-c,\ 1{+}2c,\ -c)$ in each direction has the response $1 + 4c\sin^2(k\Delta x/2)$, and $c = s^2/2$ cancels the second moment, so that the density of a long wave is correct to fourth order in the cell size. A five-point pass with the response $1 + 4c\,\sigma + c_2\sigma^2$, $\sigma = \sin^2(k\Delta x/2)$, cancels the next order as well; the inverse window of the cubic cloud has $c = 1/6$ and $c_2 = 7/15$. The pass acts on the densities of Eq. (3), before the division by the volume element. The particles carry the density per coordinate volume, and on scales near the Hubble radius most of its contrast is the contrast of the volume element itself. The division is local, so a pass after it leaves that part of the smoothing in the sources. The pass also amplifies the sampling error of the deposit at the shortest waves, which limits its use once the flow is no longer smooth (Sec. IV J).

C. Metric at the particles

The pusher of Ref. [6] reconstructs the metric and its gradient at a particle from the sixty-four surrounding cell values with cubic Hermite polynomials whose nodal slopes are central differences. The gradient of this reconstruction is accurate to second order, and its error depends on the position of the particle within the cell: for a wave of wave number k it is low by $(k\Delta x)^2/6$ at the cell centers and high by $(k\Delta x)^2/12$ midway between them. Particles on a lattice all sit at the same positions within their cells, so the force on every one of them is wrong by the same fraction, which shifts the growth rate of the wave, and the fraction changes as the particles cross cells. The reconstruction is replaced here by the cubic spline through the cell values: the weights of the cubic B-spline applied to values that are first sharpened with the inverse of the sampled spline, a five-point pass in each direction. The stencil is the same, the interpolant is twice continuously differentiable, and the error of its gradient is of fourth order and independent of the position within the cell.

D. Time step and constraint damping

A fixed fraction of a cell is a stable step at all epochs, but its accuracy is set at the start of a run. The homogeneous solution changes on the expansion time $1/\mathcal{H}$, with $\mathcal{H} = \dot{a}/a$, and the integrator of the spacetime solver is of fourth order. Its error on the homogeneous background is committed in the first steps and stays in the solution as a uniform violation of the Hamiltonian constraint, which the expansion does not dilute. In a box several Hubble radii wide the Courant step is a sizeable fraction of the initial expansion time. The step is therefore limited to a

fraction f of the expansion time,

$$\Delta t \leq \frac{3f}{|\langle \alpha K \rangle|}, \tag{5}$$

with the mean over the mesh. The limit acts only while the expansion is fast, and the extra steps grow with the logarithm of the expansion over which it acts.

The constraint damping of the Z4c formulation acts at the rate $\kappa_1 \alpha$ per unit of coordinate time. With $\alpha = a$ a constant coefficient κ_1 becomes stiff as the universe expands. The coefficient is divided by the lapse here, which damps the constraints at a fixed rate per unit of conformal time.

E. Initial data on the three-torus

The initial slice must satisfy the Hamiltonian and momentum constraints. Following Refs. [1, 2], the data start from a Newtonian density contrast δ_N and velocity v^i on a background with $a = 1$, with the potential ψ_N from $\partial^i \partial_i \psi_N = 4\pi \rho_0 \delta_N$, and from the dictionary of Refs. [7, 8], which reads them as a metric and a matter distribution,

$$\begin{aligned} \mathrm{d}s^2 &= a^2 \left[-(1 + 2\psi_N)\, \mathrm{d}t^2 + (1 - 2\psi_N)\, \delta_{ij}\, \mathrm{d}x^i \mathrm{d}x^j \right], \\ \frac{\rho}{\rho_0} &= 1 + \delta_N - 2\psi_N - 2\, \frac{\mathrm{d}\psi_N}{\mathrm{d} \ln a}, \\ K &= -3H \left(1 - \psi_N - \frac{1}{H} \frac{\mathrm{d}\psi_N}{\mathrm{d}t} \right). \end{aligned} \tag{6}$$

The source papers solve the conformal thin-sandwich equations for the corrections beyond linear order with the density held fixed. Here the particles are held fixed instead. The spatial metric is conformally flat, $\gamma_{ij} = \psi^4 \delta_{ij}$, and the extrinsic curvature is

$$\begin{aligned} K_{ij} &= \psi^{-2} \hat{A}_{ij} + \tfrac{1}{3} \psi^4 \delta_{ij} K, \\ \hat{A}_{ij} &= \partial_i W_j + \partial_j W_i - \tfrac{2}{3} \delta_{ij} \partial_k W^k, \end{aligned} \tag{7}$$

with K from the dictionary and a vector potential W^i. The particle sums that the code deposits, $\hat{E} = \psi^6 E$ and $\hat{S}_i = \psi^6 S_i$ for the energy density E and momentum density S_i seen by normal observers, are fixed by the particle positions, masses, and momenta, up to the dependence of the Lorentz factor on ψ. The constraints become

$$\begin{aligned} \partial^j \partial_j \psi &= \tfrac{1}{12} K^2 \psi^5 - \tfrac{1}{8} \psi^{-7} \hat{A}_{ij} \hat{A}^{ij} - 2\pi \psi^{-1} \hat{E}, \\ \partial^j \hat{A}_{ij} &= \tfrac{2}{3} \psi^6 \partial_i K + 8\pi \hat{S}_i. \end{aligned} \tag{8}$$

With the density scaling as the particles require, the linearized first equation has the operator $\partial^j \partial_j - \frac{9}{2} H^2$ on the homogeneous background, which is invertible on the torus; with the density held fixed the constant mode is marginal. The equations are solved on the mesh of the run with Fourier transforms, with the deposit of the code reproduced, so that the data satisfy the constraints for the sources the code will see.

The particles start on a lattice aligned with the mesh, with the lattice points at the cell corners, and are displaced by the Zel'dovich approximation exactly as in the Newtonian reference. The mass per coordinate volume must equal $\rho W \psi^6$ with Lorentz factor W, whose contrast is $\delta_N - 5\psi_N$ to linear order for growing-mode velocities; the particle masses carry the part of it that the displacement leaves out. For cold matter only the density and velocity fields enter the evolution, so the division of the mass among the particles is free. Identical positions give the two calculations the same sampling error of the deposit, which Sec. IV B shows to matter.

F. Newtonian reference

The Newtonian reference is a particle-mesh code on the mesh of the relativistic run with the same cloud shape, so that the two calculations differ in the gravity. Its output is turned into observer quantities with the dictionary above, as in Ref. [2], and light rays are integrated in the metric of the dictionary alongside the particles.

G. Observables

The comparisons use quantities defined by observers [1, 2]: the density $\rho(\tau)$ along the worldline of an observer comoving with the matter, expressed through

$$a_p = \left(1 + \tfrac{3}{2} H_0 \tau\right)^{2/3}, \qquad \delta_{\mathrm{obs}} = \frac{\rho}{\rho_0}\, a_p^3 - 1, \tag{9}$$

the proper time at which fiducial observers cross the overdensity, and the redshift and luminosity distance along light rays between the overdensity and the underdensity. For a ray with wave vector k^a, emitted and received by observers with four-velocity u^a that move with the matter,

$$1 + z = \frac{(u_a k^a)_{\mathrm{emitted}}}{(u_a k^a)_{\mathrm{received}}}. \tag{10}$$

A bundle of rays that leaves the emitter within the solid angle $\mathrm{d}\Omega$ has the cross section $\mathrm{d}A$ where it is received, and the luminosity distance is

$$D_L = (1 + z) \sqrt{\frac{\mathrm{d}A}{\mathrm{d}\Omega}}. \tag{11}$$

Each bundle here is a central ray and four neighbors at a small opening angle; the cross section follows from the separations of the neighbors orthogonal to the central ray, and the solid angle from their directions in the frame of the emitter.

IV. RESULTS

A. Homogeneous expansion

A lattice of particles at rest in a periodic box of four Hubble radii must reproduce the homogeneous solution of Sec. II. The fields have no spatial dependence, so the test isolates the time integration of the coupled system. Figure 1 compares the two couplings of Sec. III B over a fourfold expansion. With the sources held fixed during a step, the lapse, the density, and the proper time of an observer drift from the exact solution at the percent level and the Hamiltonian constraint is violated at a tenth of its largest term. With the densities converted at each stage the errors fall by four to five orders of magnitude, to the level of the fourth-order time integrator, and the constraint stays satisfied.

B. Initial data and the sampling error of a lattice

The constraint solve of Sec. III E must return the dictionary to linear order. Figure 2 shows that it does so only for a suitable lattice. A particle at a cell center sits at the cusp of the cloud-in-cell weight. Where the displacement changes sign, at the density peak and at the trough, the two neighboring particles leak into each other's cells, the deposited contrast is half again too large there, and the conformal factor inherits the excess. With the lattice points at the cell corners, each particle is shared equally by eight cells, the deposit responds linearly to a small displacement, and the solved conformal factor agrees with the dictionary, with a remainder that is second order in the amplitude.

The same mechanism acts at any amplitude. A lattice whose spacing is compressed by the fraction c no longer has its harmonics on the zeros of the cloud shape, and each harmonic leaks into the deposit with the power of c set by the smoothness of the shape. On the mesh the leakage appears at the long wavelength h/c for cells of size h, which a filter of a few cells leaves in place, and it is largest where a particle meets a kink of the shape. Figure 3 shows the largest error of the deposited density for a uniform density sampled by a compressed lattice, for clouds that are B-splines of degree one to three. The error follows

$$\frac{c}{2}, \qquad 0.4\,c^2, \qquad 0.4\,c^3 \tag{12}$$

for the linear, quadratic, and cubic cloud with one particle per cell, and more particles per cell lower and cap it. With the linear cloud of Ref. [6] and one particle per cell, the density of a cold lattice is therefore wrong at the level of the compression itself, well before the collapse. A lattice point on a cell corner is a special case, with a deficit of c^2 per direction; the two density probes of the collapse problem sit on such points. A relativistic lattice displaced so that equal masses give the coordinate density is compressed four times more than the Newtonian one for the wavelength used here. This is the reason for the identical positions of Sec. III E, and the reason for the smoother clouds added to the code.

C. Linear growth

For a small amplitude both calculations must follow linear theory. With the single-wavelength data of Ref. [1], $\delta_N = \sum_i \bar{\delta}_i \sin(kx^i)$ with $\sum_i \bar{\delta}_i = 3\bar{\delta}$, the growing mode has a constant potential, and the observer density contrast at the two extrema is

$$\delta_{\rm obs} = \pm 3\bar{\delta}\left[a_p + 3\left(\frac{H_0}{k}\right)^2 a_p^{-3/2}\right]. \tag{13}$$

The first term is the growing mode. The second is what remains of the density offset $-2\psi_N$ of the dictionary once the slower clock of an observer inside the potential is taken into account; it decays. Figure 4 shows that the relativistic run, which starts from constraint-solved data and evolves in the slicing of Sec. II, and the Newtonian reference both follow this law over two decades of expansion, at both extrema, to within the size of the second-order terms. The lapse condition keeps the slices close to those of the Newtonian gauge in this regime: with zero shift it holds $\alpha\,\gamma^{-1/6}$ fixed in time, as the dictionary metric does for a constant potential.

D. Light rays in the homogeneous universe

In the homogeneous universe a ray emitted at the Hubble rate H_e has

$$D_L = \frac{2}{H_e}\,(1+z)^2\left(\sqrt{1+z}-1\right). \tag{14}$$

Figure 5 compares bundles emitted at two times with this relation. The redshift agrees to the accuracy of the background. The luminosity distance is low by a small amount that falls along the ray, and that amount is the second-order time error of the pusher: the metric at the middle of a step is the mean of the two time levels, which lowers the coordinate speed of light by $(\mathcal{H}\Delta t)^2/8$ for the conformal Hubble rate $\mathcal{H}$ and time step Δt. The measured error coincides with this expression along every ray, so the launch, the cross section, and the solid angle of the bundles are right.

E. Collapse at one wavelength: general relativity against Newtonian gravity

With the two calculations started from identical particle positions on the same mesh, with the same cloud shape and time step, the comparison isolates gravity.

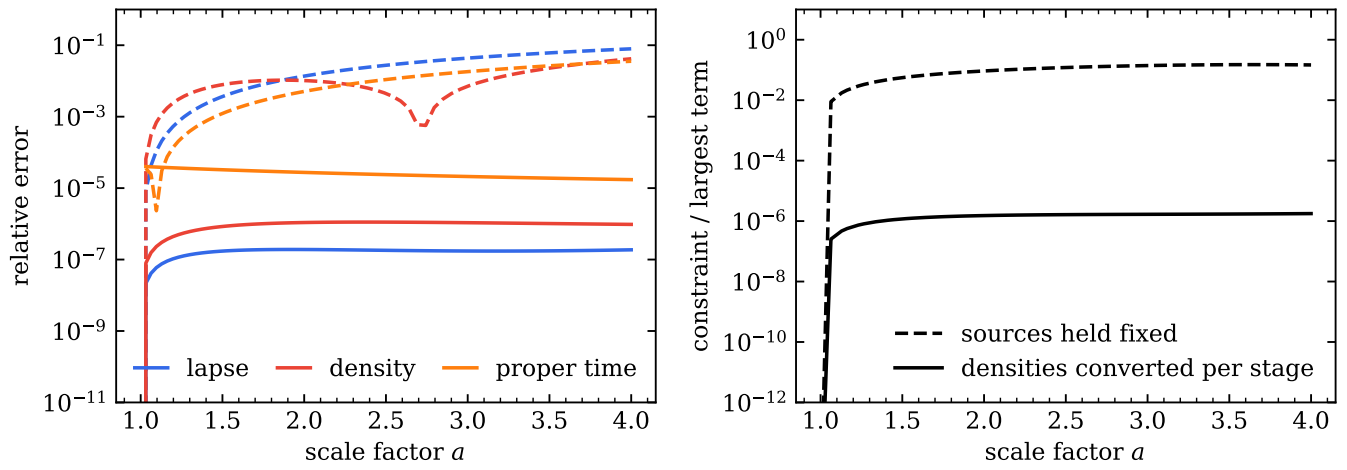


FIG. 1. Homogeneous universe. With the matter sources held fixed during a spacetime step (dashed), the lapse, the density, and the proper time of an observer drift from the exact solution; with the densities converted at each stage (solid) the errors are those of the time integrator. Left: relative errors against the exact matter-dominated solution; the proper volume of the box has the error of the density. Right: root mean square of the Hamiltonian constraint over the box in units of its largest term, $16\pi\rho$.

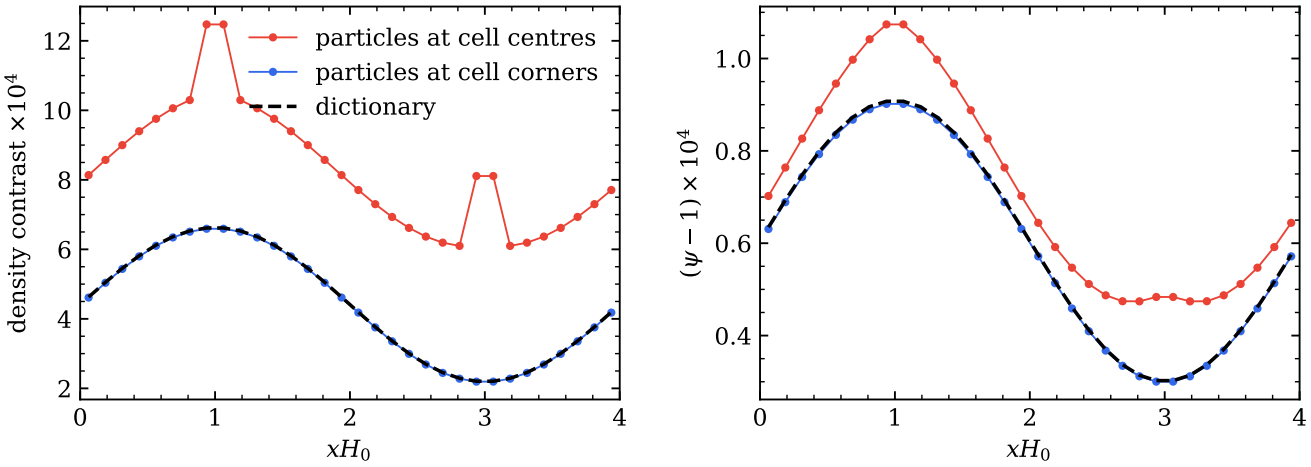


FIG. 2. Initial data at small amplitude along the mesh line next to the density peak. Particles at the cell centers (red) deposit a density contrast far above the dictionary value (dashed), with spikes where the displacement changes sign, and the solved conformal factor inherits the excess. Particles at the cell corners (blue) reproduce the dictionary. Left: deposited density contrast. Right: conformal factor.

Figure 6 shows the observer density contrast at the two extrema for the four amplitudes of Ref. [1], on a mesh of sixty-four cells per wavelength with one particle per cell. At the underdensity the two calculations coincide for the three lower amplitudes, as in Ref. [1], and separate only for the largest. At the overdensity the contrast rises steeply when the matter streams cross, and the rise comes later in the relativistic calculation, by an amount that grows with the amplitude.

The proper time at which a fiducial observer reaches the overdensity measures the delay without reference to coordinates [1]. Figure 7 follows observers that start at four distances from the overdensity. In both calculations they fall in, cross, and oscillate about the bound region, and the relativistic infall is later. Figure 8 shows the delay, the ratio of the proper-time scale factors at which the distance has fallen to a tenth of its initial value in the two calculations, against the amplitude. The delay is proportional to the amplitude over the whole range, the extreme case included. It is the same for the four starting distances and for the three cloud shapes of Sec. IV B. It is also the same when measured in coordinate time, and the clocks of the observers agree between the two calculations, so the infall itself is later. The dictionary that turns the Newtonian calculation into a spacetime is exact to first order in the perturbation, and the wavelength is

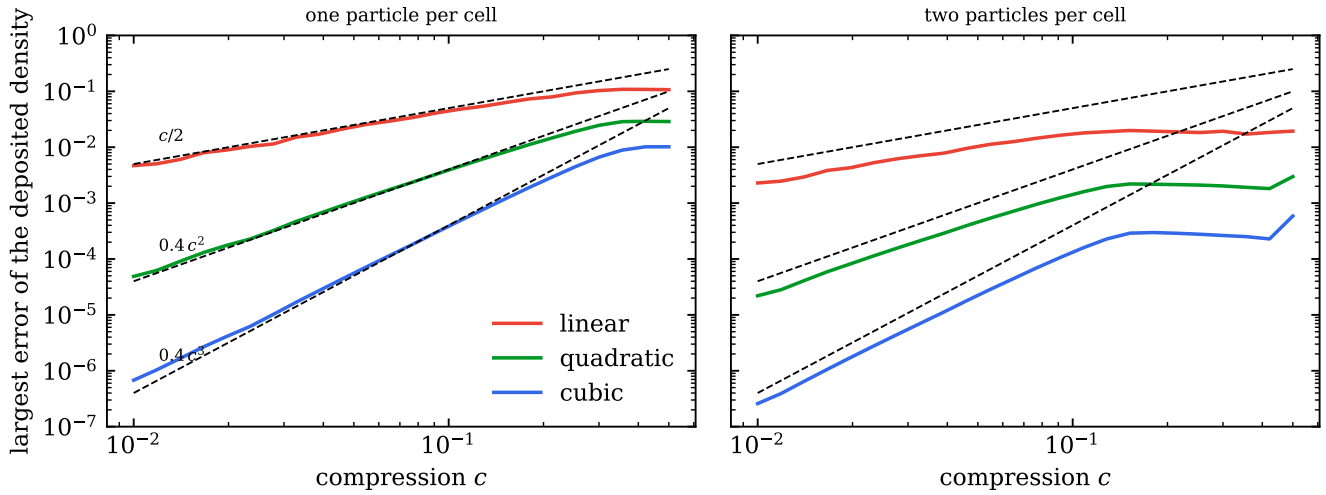


FIG. 3. Sampling error of a cold lattice. The largest error of the deposited density, for a uniform density sampled by a lattice compressed by the fraction c, falls with the smoothness of the particle cloud: it follows $c/2$, $0.4\,c^2$, and $0.4\,c^3$ (dashed) for the linear, quadratic, and cubic cloud with one particle per cell (left). Two particles per cell and direction lower and cap it (right). The deposit is filtered as in the runs.

several Hubble radii at the start; a difference of second order, proportional to the amplitude in relative terms, is what the comparison can show, and it shows one.

The delays read from the figures of Ref. [1] are larger for the three lower amplitudes and depend weakly on the amplitude. Its Newtonian reference had a finer force resolution than its relativistic code, and its text notes that finite resolution delays the collapse. With the same resolution in both calculations the delay is that of gravity alone for the given initial data.

The Newtonian calculation has no scale, so its infall does not depend on the wavelength; the relativistic one does. Repeating the comparison at one amplitude with the wavelength halved and doubled changes the delay by about a factor of four each way (Fig. 8, right): the delay follows the depth of the initial potential at the overdensity, about three times that depth, across all amplitudes and wavelengths run. It belongs to the horizon scale and vanishes for perturbations well inside the horizon, where the Newtonian calculation with the dictionary is accurate. At the longest wavelength the potential is as deep as for the extreme amplitude, and both calculations end near the collapse in the same way. How the delay divides between the second-order part of the initial data and that of the evolution is open: the nonlinear part of the constraint solve lowers the initial contrast at the overdensity by a fraction proportional to the potential, in the direction of a later collapse.

At the largest amplitude the lapse at the overdensity turns around and collapses while the density there rises without bound, and the run ends; the Newtonian dictionary fails slightly earlier, where its potential reaches one half. Reference [1] reports the same end for both calculations and a black hole forming. The lapse condition of Sec. II holds $\alpha\,\gamma^{-1/6}$ fixed, so the lapse vanishes only with the volume element. A first alternative that holds $\alpha - \gamma^{1/6}$ fixed, and so stops the slices at a finite volume element, ends at the same place: the lapse then falls only as fast as the local scale factor. A driver toward constant mean curvature, which adds to the lapse equation a term proportional to the difference between the local K and its mean over the box, does what is needed: it vanishes in the homogeneous universe, leaves the observer quantities of the linear regime unchanged, and at the largest amplitude brings the lapse at the overdensity down to a few percent of its largest value while the density there grows slowly. That run ends later, when the collapsing region, a few cells across, is no longer resolved. With three levels of static refinement around the overdensity, onto which the initial data are interpolated, the collapse is followed to smaller scales and the lapse falls to a tenth of its largest value, but the spatial metric then degenerates in the innermost cells; a first shift driver, added against the stretching of the coordinates into the collapsing region, is itself unstable on the refined blocks. Following the black hole needs a shift condition that is stable there and initial data that satisfy the constraints on the refined blocks. This is the state of the extension.

F. Dependence on resolution

Reference [1] finds that the collapse comes earlier with resolution in both calculations and that the error at late times is set by the number of particles. Figure 9 repeats that study at one amplitude with the quadratic cloud and one particle per cell. The infall of the fiducial observers moves earlier by about a percent from sixty-four to ninety-six cells per wavelength and by a few parts in a thousand from there to one hundred and twenty-eight, in

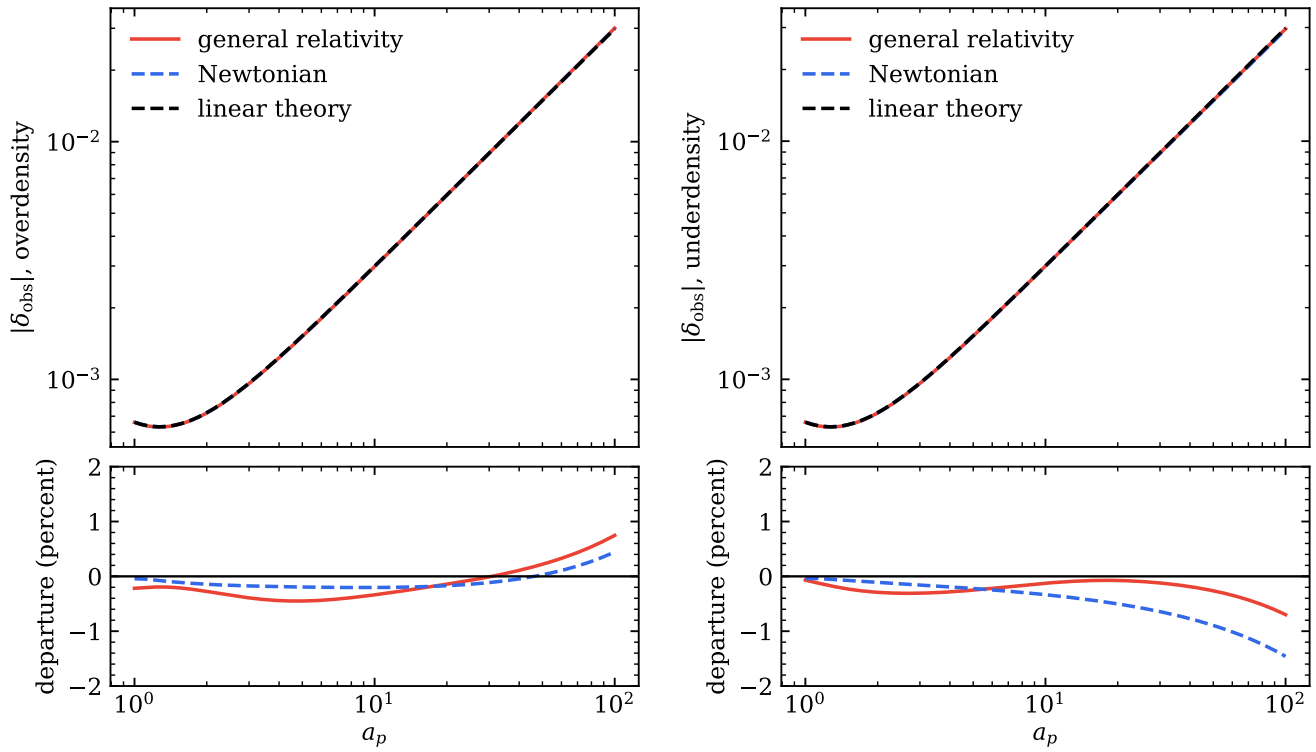


FIG. 4. Linear regime. The observer density contrast of the relativistic run (red) and of the Newtonian reference (blue, dashed) follows linear theory (black, dashed) over two decades of expansion at the overdensity (left) and the underdensity (right). Lower panels: departure from linear theory, which includes the smoothing of the density probe.

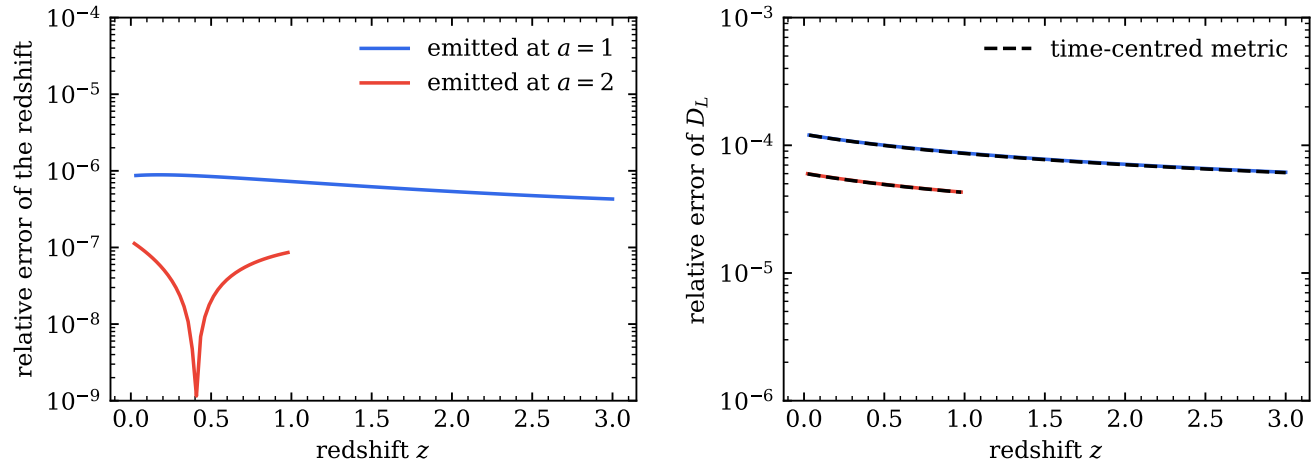


FIG. 5. Light rays in the homogeneous universe. The redshift along bundles emitted at two times agrees with the exact relation (left), and the error of the luminosity distance (right) coincides with the one expected of a pusher that centers the metric in time (dashed).

both calculations alike, and the relativistic delay changes by less than a fifth of a percentage point. Replacing the metric reconstruction at the particles by the spline of Sec. III C moves the relativistic infall by less than a thousandth: the wave is well resolved here. In the production configuration of Sec. IV J, applied to both calculations, the infall times at sixty-four cells equal those of the quadratic cloud at one hundred and twenty-eight to a few thousandths, and the delay is the same. The density reached in the bound region rises with resolution, as in Ref. [1]. The mean relative Hamiltonian constraint converges while the lattice is nearly undeformed and then

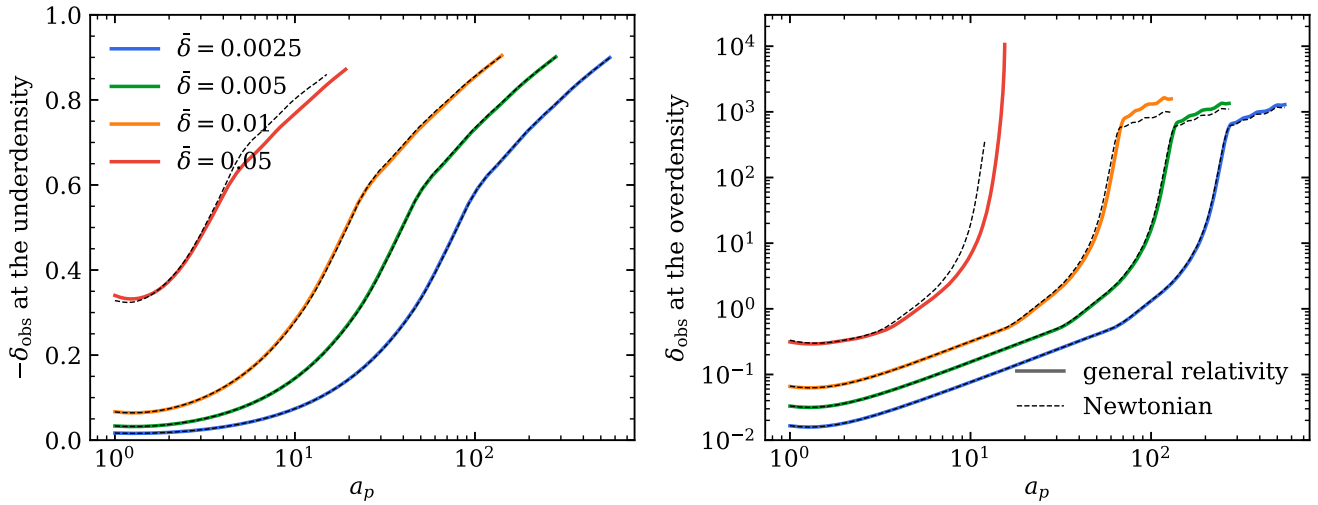


FIG. 6. Collapse at one wavelength. The observer density contrast of the relativistic run (colors) and of the Newtonian reference (black, dashed) coincides at the underdensity (left) for the three lower amplitudes; at the overdensity (right) the relativistic rise comes later, increasingly so for larger amplitude. The relativistic run at the largest amplitude ends when the lapse collapses at the overdensity, the Newtonian one where its potential reaches one half.

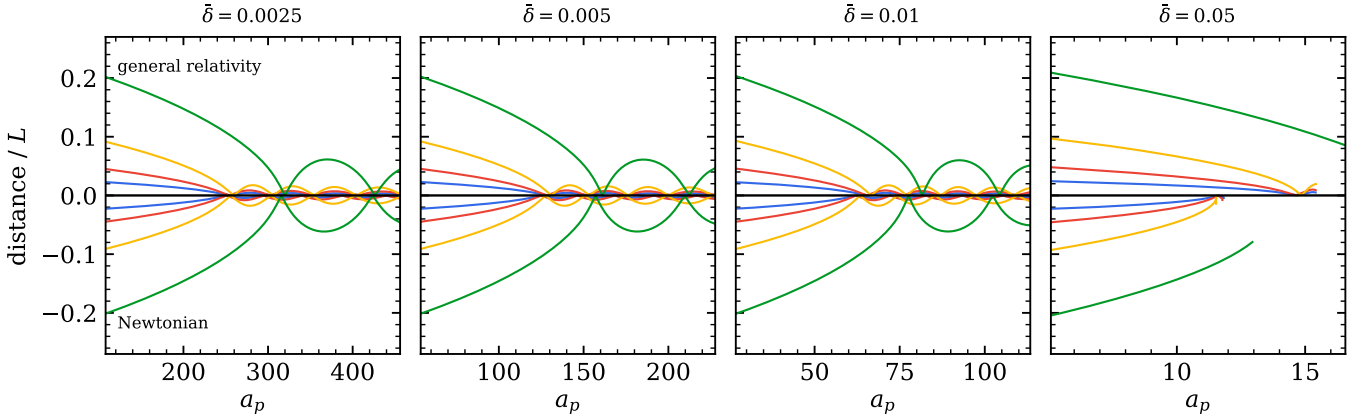


FIG. 7. Fiducial observers fall into the overdensity later in the relativistic run (upper half of each panel) than in the Newtonian reference (lower half, mirrored). Coordinate distance from the overdensity, in units of the wavelength L, of observers that start at $L/32$, $L/16$, $L/8$, and $L/4$ along one axis, against their own proper-time scale factor. At the largest amplitude both calculations end during the first infall.

becomes independent of the cell size: at one particle per cell the sampling error of Sec. IV B depends on the compression of the lattice and not on the mesh.

G. Constraint violation and the cloud shape

The collapse runs measure how the cloud shape acts on the constraints. Following Ref. [3], a constraint is reported relative to the root of the sum of the squares of its terms, H_r for the Hamiltonian and M_r for the momentum constraint. Figure 10 shows their means over the mesh. While the flow is single-valued the violation is set by the deposit. The deposited fields satisfy the continuity equations exactly as functions of position, and the finite differences of the spacetime solver see them through the mesh, so the mismatch is the part of the cloud that the mesh does not resolve. The quadratic cloud lowers the violation by one to two orders of magnitude and the cubic cloud by more, down to the level of the time integration at early times. Once the bound region has formed, the three shapes agree: the matter there has many velocities within a cell. The smoother clouds also remove a bias of the collapse itself. The linear cloud holds the central density down, and the inner observers fall in several percent later than with the smoother clouds, in both calculations alike; the relativistic delay of Fig. 8 is unaffected.

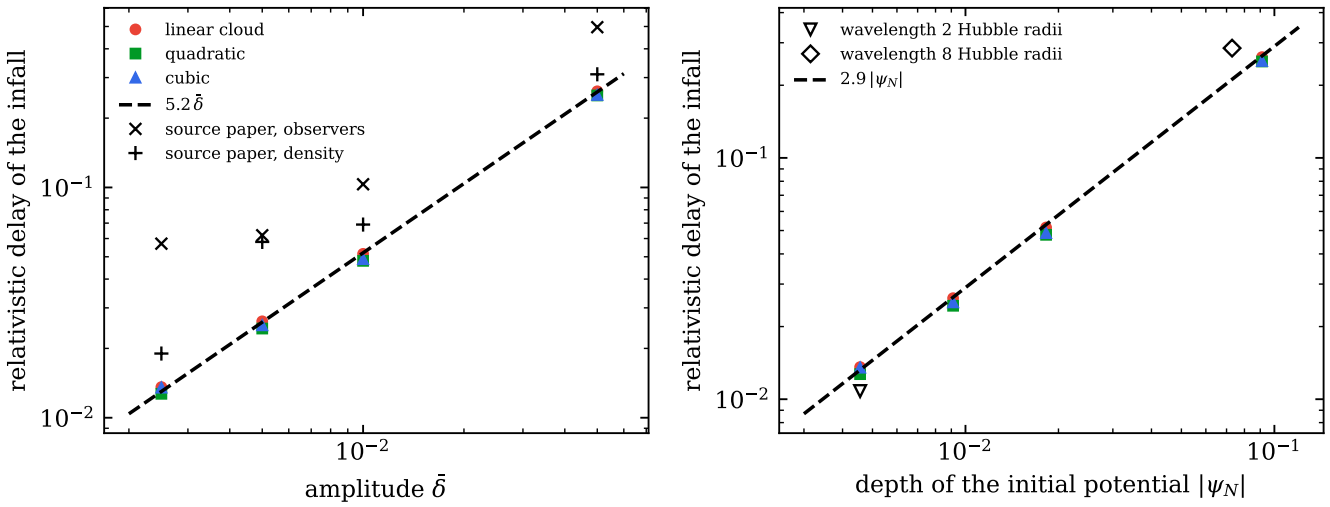


FIG. 8. The relativistic delay of the infall is set by the depth of the initial potential. Left: ratio, minus one, of the proper-time scale factors at which the distance of a fiducial observer has fallen to a tenth of its start in the relativistic and in the Newtonian calculation, averaged over the starting distances, against the amplitude, for the three cloud shapes; dashed: $5.2\,\bar{\delta}$; crosses: the same ratio and the ratio at a density contrast of ten, read from the figures of Ref. [1]. Right: the same delay against the depth of the initial potential at the overdensity, with two further wavelengths at the amplitude 0.01 (open symbols); dashed: $2.9\,|\psi_N|$.

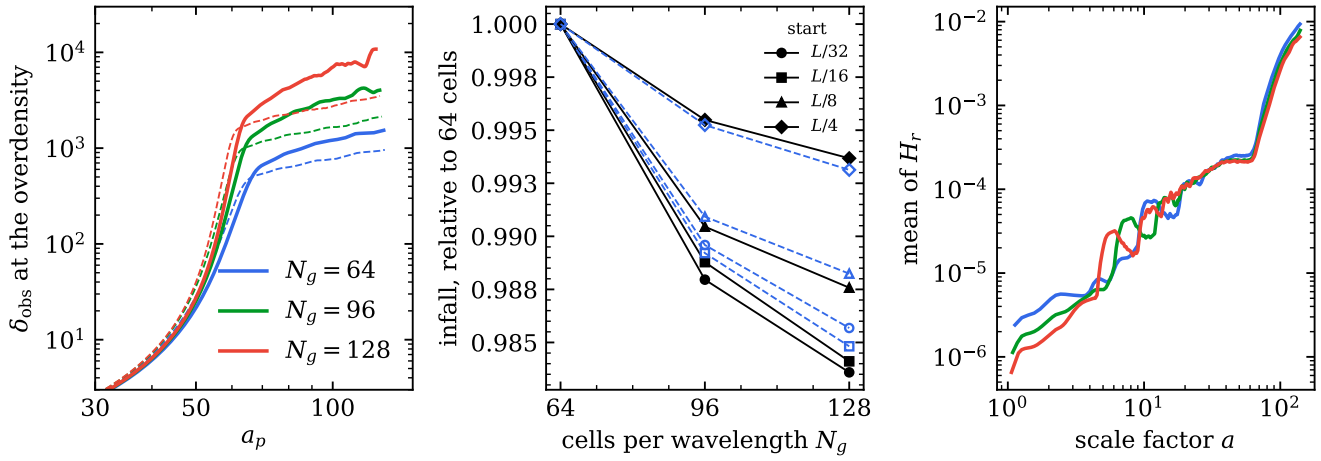


FIG. 9. Resolution study at the amplitude 0.01 with the quadratic cloud. Left: observer density contrast at the overdensity, relativistic (solid) and Newtonian (dashed); the density of the bound region rises with resolution. Center: scale factor of the infall of the fiducial observers relative to its value at 64 cells, relativistic (black) and Newtonian (blue, dashed), for the four starting distances. Right: the mean relative Hamiltonian constraint converges early and is set by the particle sampling later.

H. Light rays through the collapsing universe

Light probes the spacetime along its path. Bundles of rays leave the overdensity toward the underdensity, and the reverse, at a set of emission times; the Newtonian bundles leave at the same proper times of the emitters and are integrated in the metric of the dictionary. Figure 11 shows the luminosity distance along each ray relative to the homogeneous relation of Sec. IV D for an emission at the proper time of the emitter. Rays emitted while the perturbation is linear deviate from that relation at the level of the perturbation, with opposite signs for the two directions. Rays that leave the overdensity after it has collapsed gain distance with little redshift at first, and rays from the underdensity are blueshifted as they fall into the bound region, where their redshift turns back. Both calculations show the same structure, and the deviations agree with those read from the figures of Ref. [1] to about a tenth of their size. The two theories of gravity differ far less than either differs from the homogeneous universe: by less than a part in a thousand for the rays emitted in the linear phase, and at the per-

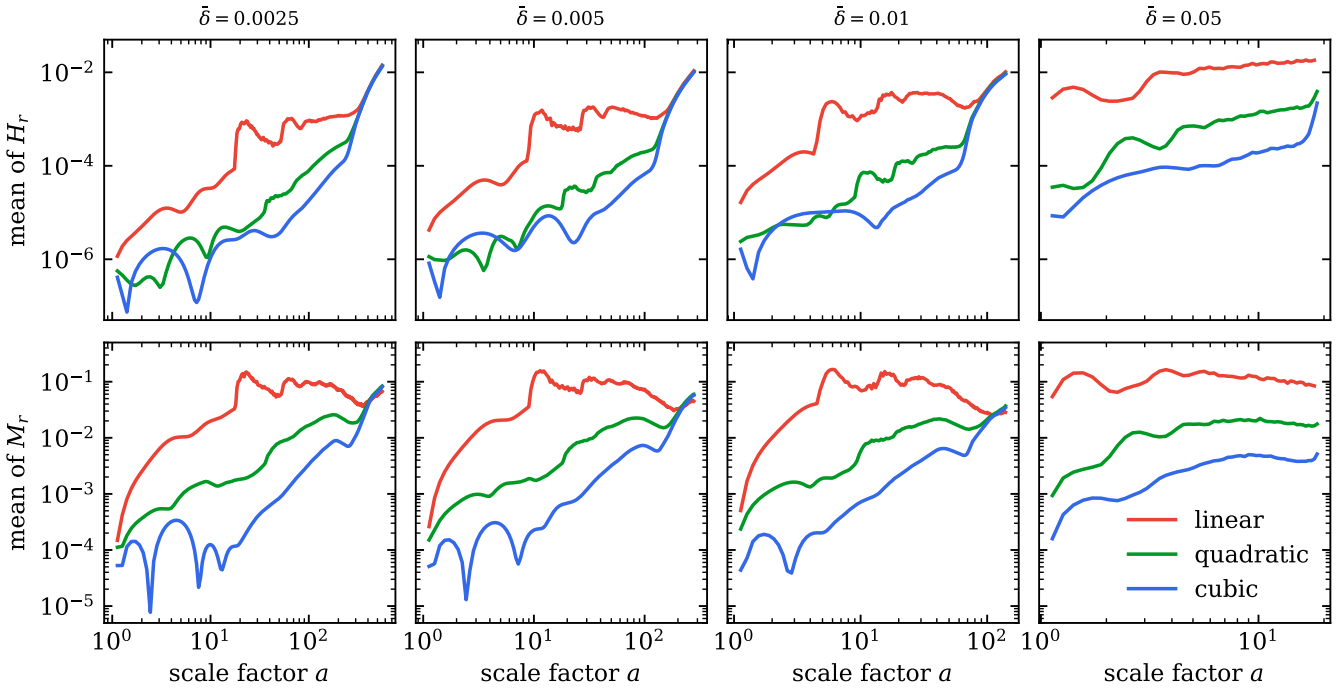


FIG. 10. A smoother particle cloud lowers the constraint violation of the collapse runs until the matter streams cross. Mean over the mesh of the relative Hamiltonian constraint (top) and momentum constraint (bottom) against the scale factor of the background, for the four amplitudes and the linear, quadratic, and cubic cloud.

cent level only where a ray starts or ends in the bound region, the conclusion of Ref. [1].

I. Time integration over a thousandfold expansion

Daverio, Dirian, and Mitsou test the time integration of a relativistic particle code from redshift one thousand to today [3]. Their tests are repeated here with their box sizes, cell numbers, and noise amplitudes, in the units of this code.

In the first test the universe is homogeneous. With exact data the errors are those of the time integrator and converge at fourth order (Fig. 12, left). They are larger than in Ref. [3] and grow through the run. Part of this is the constraint variable Θ of the Z4c formulation, which enters the lapse condition and lets the lapse drift off conformal time while it decays; damping the constraints at a fixed rate per unit of conformal time, which amounts to dividing the damping coefficient by the lapse, removes that part. The rest is the truncation error of the first steps, which fixes the normalization of conformal time for the remainder of the run. With white noise on every field, the second test, the evolution is stable at all three resolutions and the largest constraint violations fall by seven orders of magnitude during the expansion, as in Ref. [3] (Fig. 12, center and right). The density error of the noisy runs grows more slowly than the scale factor, as required. Its level depends on the momentum noise of the particles: with the amplitude stated in Ref. [3] the particles drift freely by enough to exceed the density error reported there before any gravitational growth, and without momentum noise the early error equals the reported one. The runs of Ref. [3] are therefore matched by the second reading.

In the third test a plane wave of small amplitude starts outside the horizon and ends well inside it. The potential must stay constant and the density contrast must follow linear theory. Figure 13 shows both: the potential changes by a fraction that falls by four per doubling of the cell number, and the departure of the density from linear theory converges at second order, below the values of Ref. [3] at equal cell number. The relative Hamiltonian constraint falls with resolution, where Ref. [3] reports it growing.

J. A random field from a power spectrum

The fourth test of Ref. [3] starts at redshift one thousand from a Gaussian random field with a cosmological power spectrum, with all power below a cut-off wavelength removed, in two boxes. In the larger box the field stays linear to the end. In the smaller box the highest peaks collapse before the end. The initial data are built from the field as in Sec. III E. Ref. [3] does not specify its spectrum; a fitting formula with current parameters is used here, and one realization serves all resolutions.

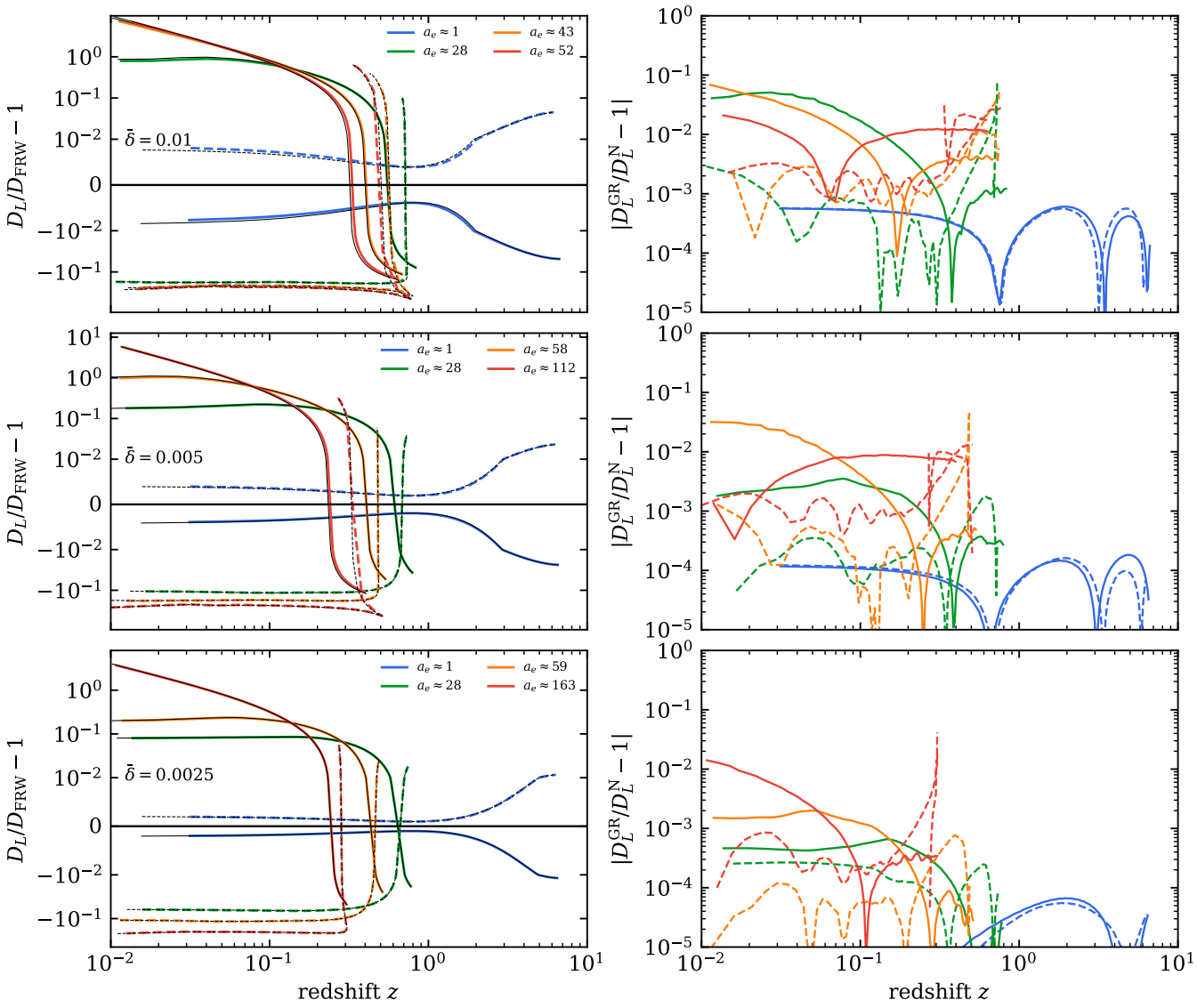


FIG. 11. Light rays between the overdensity and the underdensity deviate strongly from the homogeneous relation, and nearly equally in the two theories of gravity. Left: luminosity distance over the homogeneous relation, minus one, against the redshift, for rays emitted from the overdensity (solid) and from the underdensity (dashed) at the scale factors given; colors: relativistic run; thin black: Newtonian reference. Right: ratio of the relativistic to the Newtonian luminosity distance at equal redshift, minus one. The vertical axis on the left is linear between $\pm 10^{-2}$.

Figure 14 shows the larger box. Five properties of the method set the errors, and the test separates them. The first is the step. With the Courant step alone the Hamiltonian constraint is a uniform offset that falls by sixteen per halving of the step: the error of the integrator on the background (Sec. III D). With the step limit the offset is absent. The second and third are the cloud and the metric interpolation, which together fix the self-gravity of a wave. Without the compensation of Sec. III B, the power of the shortest initial waves, ten cells long at the coarsest resolution, falls short of linear theory by an amount that grows through the run and scales with the second moment of the cloud. With the compensation and the Hermite reconstruction the shortest waves overshoot instead, by the gradient error of that reconstruction at the lattice positions. With the compensation to fourth order and the spline of Sec. III C, every wave of the initial data follows linear theory over the thousandfold expansion, up to the growth of second order in the contrast near the end. The fourth is the order of the cloud. Once the displacements reach a fraction of the particle spacing, the sampling error of the lattice enters. With the quadratic cloud both constraints then rise by up to two orders of magnitude toward the end; with the cubic cloud

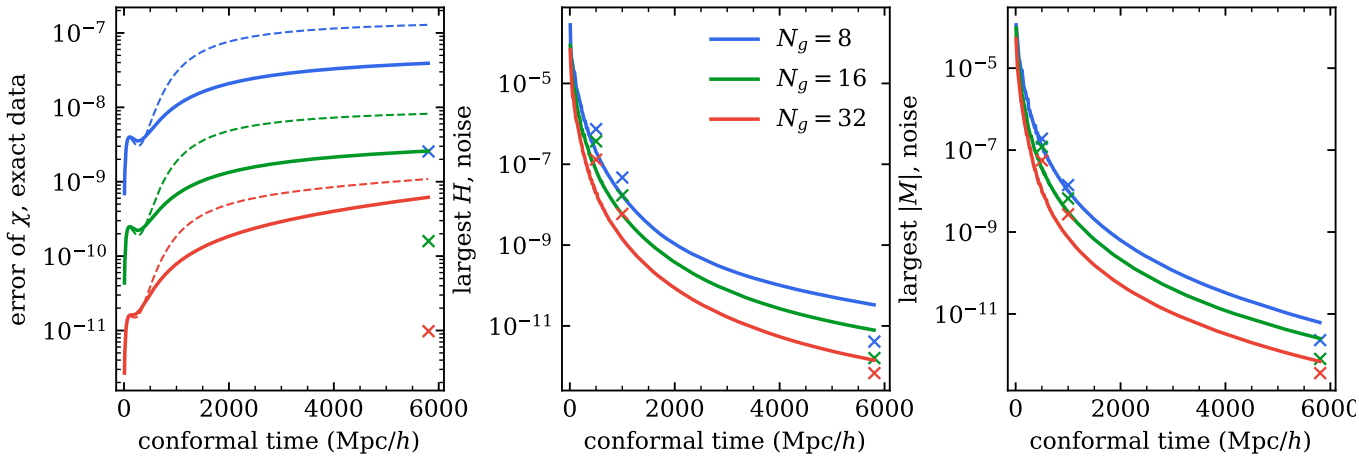


FIG. 12. Homogeneous universe from redshift one thousand to today at 8, 16, and 32 cells. Left: the error of the conformal factor for exact data converges at fourth order; dashed: constant damping coefficient; solid: damping at a fixed rate per unit of conformal time. Center and right: with white noise on every field the largest Hamiltonian and momentum constraint violations decay throughout the expansion. Crosses: values read from the figures of Ref. [3], in its units.

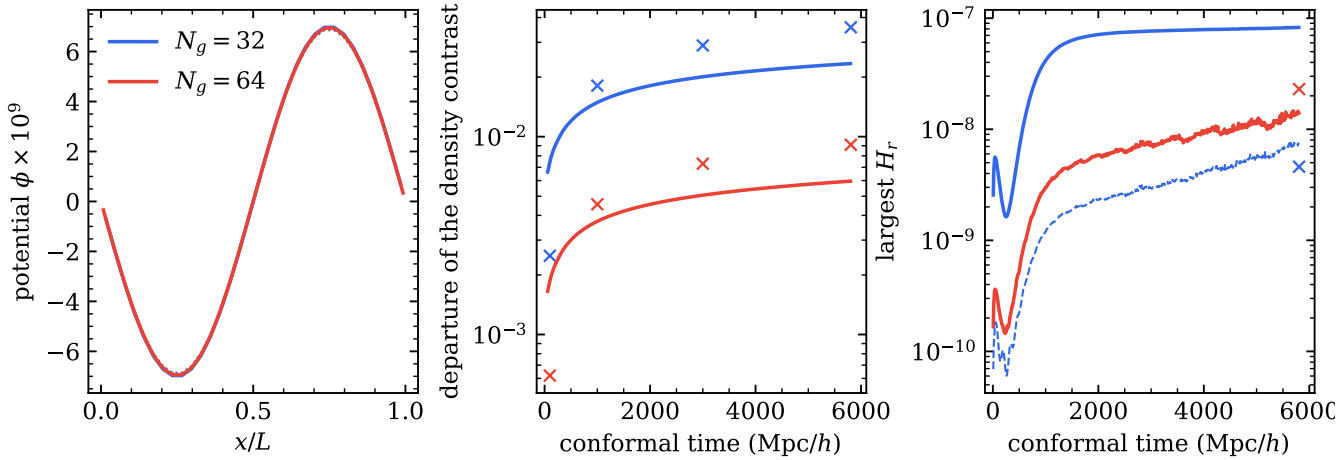


FIG. 13. Plane-wave linear mode from redshift one thousand to today. Left: the potential at the start (dotted) and at the end (solid) coincide. Center: the largest departure of the density contrast from linear theory, relative to the largest contrast, converges at second order. Right: the largest relative Hamiltonian constraint falls with resolution; dashed: 32 cells with the smaller time step of Ref. [3]. Crosses: values read from the figures of Ref. [3].

they stay within about an order of magnitude of their early values. The fifth is the resolution of the terms of second order in the contrast, which carry wave numbers up to twice the cut-off of the initial spectrum. With ten cells per cut-off wave these terms are sampled with five cells per wave, and their truncation error dominates the Hamiltonian constraint late in the run for every cloud. The constraint field shows this directly: its power lies between the cut-off and twice the cut-off, and it follows the Laplacian of the squared contrast. With twenty cells per cut-off wave both constraints stay low to the end.

The excess of the power over linear theory near the end is ordinary nonlinear growth. It is the same at 64 and 128 cells, and a Newtonian particle-mesh run of the same field, with the same particles, cloud, compensation, and force interpolation, shows the same excess over its own linear theory. The relativistic power lies below the Newtonian one by a few parts in ten thousand at all epochs, the size expected of a relativistic correction at this depth of the potential; its dependence on resolution is being measured.

Reference [3] also tests the conservation of energy and momentum of the matter on the mesh and finds residuals of a percent at every resolution. The same test here uses the density form of the equations, in which the terms of the homogeneous background cancel, so that the resid-

uals are measured against the perturbations alone. The momentum residual is the product of the response of the deposit and that of the gradient of the metric reconstruction, minus one. With the Hermite reconstruction it is a few thousandths, uniform over the mesh; with the spline and the compensation it falls with the fourth power of the cell size.

Figure 15 shows the smaller box, where the contrast reaches order one. At early times the picture is that of the larger box. In the production configuration the power follows linear theory up to the cut-off, where Ref. [3] and the runs without the new elements fall short by amounts that grow with the wave number, and the constraints are at or below those of Ref. [3] at equal cell number, the momentum constraint by an order of magnitude. The departures of the power from linear theory that then develop are converged: the runs at 64 and 128 cells agree in every shell at every epoch, to the end, where the contrast is of order one. At 64 cells the cut-off wave has ten cells, and the Hamiltonian constraint rises as the square of the contrast from early on, the truncation error of the terms of second order; at 128 cells it stays at its initial level until the contrast of the highest peaks is a few tenths. This early rise at 64 cells is the same for every cloud, for either form of the damping, and for twenty-seven particles per cell, the number of Ref. [3]. From the first collapse on, both constraints rise steeply at either resolution and end two orders of magnitude above the Hamiltonian constraint of Ref. [3] and at several times its momentum constraint. The bound regions are sampled by few particles per cell, and the compensation amplifies their noise. Without the new elements, 128 cells with one particle per cell are worse than 64 cells with eight from early on, because the sampling error of the quadratic cloud then dominates. The evolution is stable to the end in every case.

K. Initial conditions with a spectrum of scales

Barrera-Hinojosa and Li derive the initial conditions of a relativistic particle code in a constant-mean-curvature slicing and test them at the initial time [5]. The particle positions must carry the particle-number contrast of the threading. On scales of the Hubble radius it exceeds the synchronous density contrast of a Newtonian simulation by a multiple of the curvature perturbation, which is constant in time. The velocities follow either from the growth rate or from the difference of two displacement fields at neighboring redshifts. All tests of Ref. [5] concern linear theory and the generator, so they are repeated here without evolution, for matter and a cosmological constant, with the same parameters, the same two boxes, and both starting redshifts. Every transfer function has a closed form in this case, and the generator is checked against the closed forms through the gauge transformations and the constraints.

Figure 16 shows three results. The growth rate f_p of the particle-number contrast falls from the matter value on small scales to zero beyond the Hubble radius. The closed form passes through the values of Ref. [5] up to the effect of radiation, which is absent here and lowers the growth rate of Ref. [5] on small scales by the factor of Mészáros. The power P_p of the particle-number contrast of the realized initial conditions, relative to the synchronous power P_S, equals the linear ratio on the realized modes in every bin, and agrees with Ref. [5] within the spread between realizations of the few largest modes. The finite-difference velocity carries more power $P_{\theta\theta}$ of the velocity divergence than the growth-rate velocity by the truncation error of the central difference, $2\,\Delta z^2/(1+z)^2$ for a redshift offset Δz, in every bin and in both gauges, as in Ref. [5].

Two statements of Ref. [5] are refined. The scatter it reports at the smallest redshift offset is reproduced by the rounding of tabulated input spectra and is far larger than single-precision arithmetic of the displacement field would give. For particles on a lattice the window correction of a mesh estimator is replaced by the linear response of a displaced lattice, which is derived and verified here.

The lapse condition of Sec. II keeps the Newtonian slicing while the potential is constant, which holds for the growing mode in matter domination, and zero shift keeps the threading in which the particle-number contrast is defined. The initial data of Sec. III E carry that contrast: the displacement supplies the synchronous part and the particle masses the remainder, and the constraint solve supplies the metric. They are therefore the counterpart of the initial conditions of Ref. [5] for a hyperbolic code. With a cosmological constant the slicing drifts from the Newtonian one late in the evolution, by the fraction of the potential by which the growth rate falls below one.

V. METHOD DEVELOPMENTS

The first project traced most of its difficulties to one coupling: the source on the mesh is only as smooth as the local cell size and particle number allow. The cosmological tests take this coupling apart. They separate the contributions to the error of a run that starts from a particle lattice, and each but the last has its remedy.

The first is the sampling error of the lattice, which follows the compression laws of Sec. IV B. Its order in the compression is the order of the cloud. For a random field with waves down to ten cells, the density error of the cubic cloud is about a tenth of that of the quadratic cloud at equal particle number, from the first displacements until the contrast is of order one. More particles per cell help the quadratic cloud little and the linear cloud not at all before that point. Sub-particles interpolated between Lagrangian neighbors, the phase-space sheet of Ref. [9], are worse than no refinement for the quadratic and cubic clouds in this regime, because the interpolation of the displacement is itself an error of second order in the particle spacing.

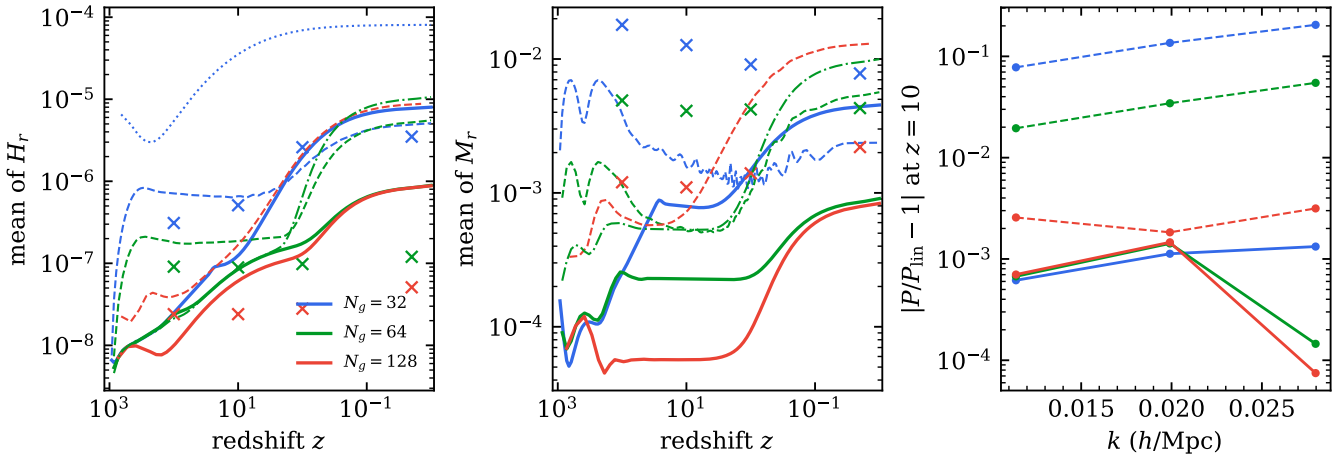


FIG. 14. Random field in the larger box of Ref. [3], linear to the end, at 32, 64, and 128 cells. Left and center: means over the mesh of the relative Hamiltonian and momentum constraints against redshift. Solid: cubic cloud, compensation to fourth order, spline interpolation of the metric, and step limit; dashed: quadratic cloud without the compensation; dash-dotted: quadratic cloud with the compensation at 64 cells; dotted: 32 cells with the Courant step alone. Right: departure of the power spectrum from linear theory at redshift ten up to the cut-off of the initial spectrum. Crosses: values read from the figures of Ref. [3].

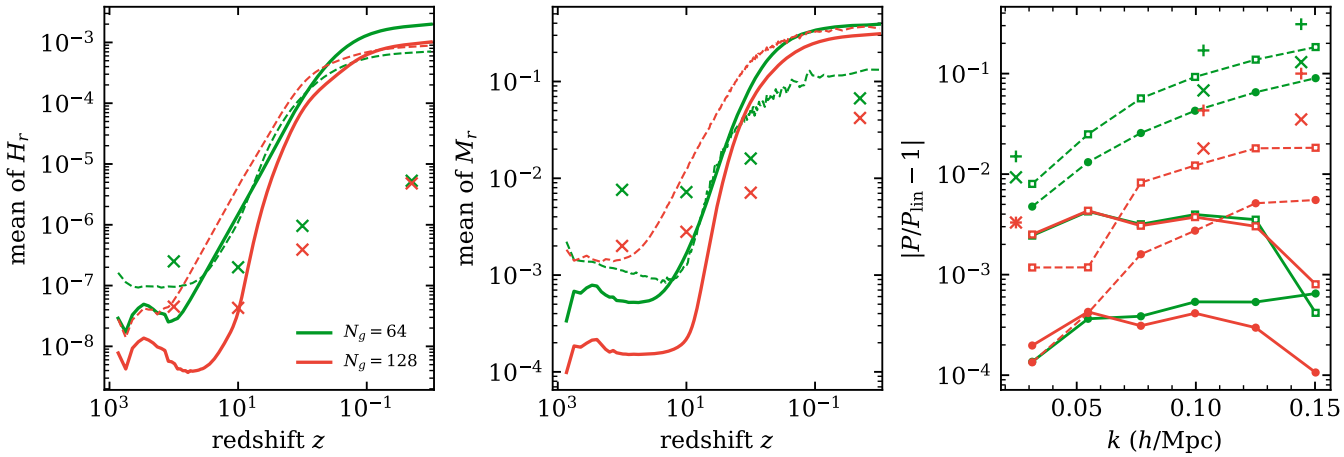


FIG. 15. Random field in the smaller box of Ref. [3], in which the highest peaks collapse before the end. Left and center: means over the mesh of the relative constraints against redshift at 64 (green) and 128 cells (red). Solid: production configuration (cubic cloud, compensation to fourth order, spline interpolation, step limit); dashed: quadratic cloud without the compensation, with eight particles per cell at 64 cells and one at 128. Right: departure of the power spectrum from linear theory at redshift 100 (filled symbols) and 10 (open symbols). Crosses and plus signs: values read from the figures of Ref. [3] at redshift 100 and 10.

The second is the smoothing mismatch. The cloud smooths the deposited density, while the force on a particle is interpolated from the unsmoothed metric. The self-gravity of a wave is then weakened by the window of the cloud, its growth is slowed, and the field equations see a force density that differs from the one that moves the particles. The compensation of Sec. III B removes this. The pass must act on the density per coordinate volume. Applied to the source after the division by the volume element it leaves a remainder that grows toward the Hubble radius, where the contrast of the volume element exceeds that of the matter; this is a relativistic effect with no counterpart in a Newtonian particle-mesh code.

The same balance has a second factor, the gradient of the metric reconstruction at the particles (Sec. III C). The Hermite reconstruction of the method makes an error of second order there that depends on where a particle sits in its cell, so that the outcome of a lattice start depends on the placement of the lattice; for one placement it nearly cancels the smoothing of the linear cloud, which hides both. The residual of the momentum equation on the mesh measures the product of the two factors directly, and with the spline and the compensation to fourth order

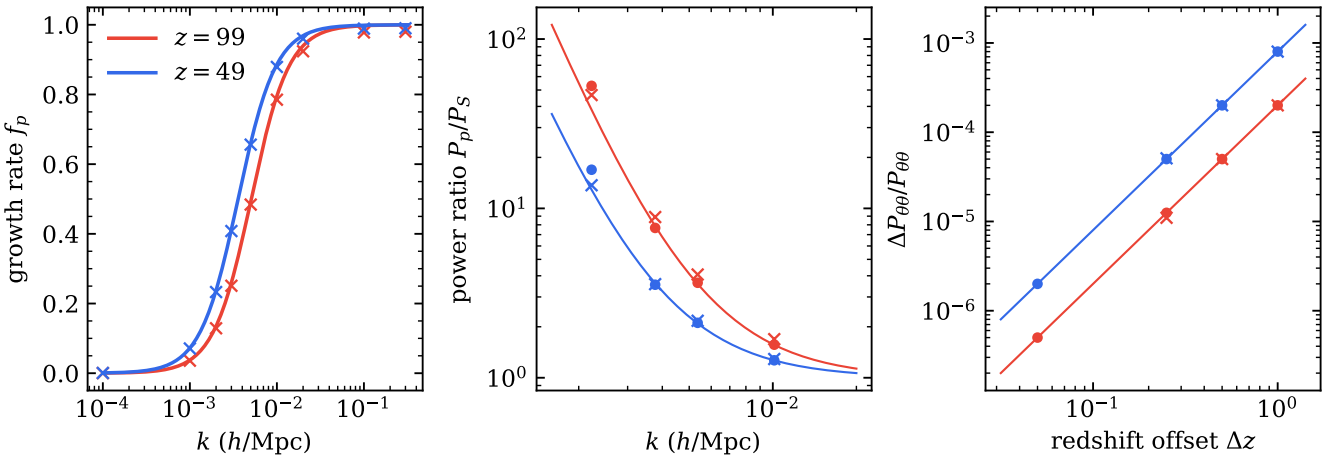


FIG. 16. Initial conditions with a spectrum of scales, at redshifts 99 and 49. Left: the growth rate f_p of the particle-number contrast vanishes beyond the Hubble radius. Center: the power of the particle-number contrast exceeds the synchronous power on the largest scales of a box of $4\,h^{-1}$Gpc; lines: linear theory at the bin center. Right: excess velocity power of the finite-difference velocity over the growth-rate velocity; lines: $2\,\Delta z^2/(1+z)^2$. Dots: this work. Crosses: values read from the figures of Ref. [5], which include radiation.

the residual, the growth of the shortest resolved waves, and the constraints of the smooth phase all improve together. A limiter that withholds the pass where neighboring cells differ strongly was tested and brings no gain.

The third is the error of the time integrator on the expanding background, which the step limit of Sec. III D removes.

The fourth is the resolution of the terms of second order in the contrast. They live at up to twice the largest wave number of the initial data, so an initial spectrum that is resolved with ten cells per wave leaves them with five. Their truncation error then sets the Hamiltonian constraint as soon as the contrast has grown, independently of the cloud, the particle number, the step, and the damping. A spectrum should therefore be cut off at twenty cells per wave.

The fifth is the noise of regions where the matter streams have crossed. There the compensation, which amplifies the shortest waves, raises the constraint violation, and the limiter does not prevent it: the amplified noise has moderate contrast and fills the collapsed regions. Lowering this noise is the purpose of the second development, the evolution of the departure of the distribution from a known solution, which applies to the equilibrium clusters of the first project and has not been started.

The constraint damping plays a minor part. A rate above the one used here raises the Hamiltonian constraint in proportion, a lower rate changes nothing, and the linear random-field run is stable to the end without damping.

Appendix A: Progress ledger

This appendix belongs to the working paper only. Table I lists the targets and their state.

Numbers behind the statements of the main text. C0, thirty-two cells, a quarter-cell step, fourfold expansion. Densities converted per stage: lapse within 2×10^{-7}, density and proper volume within 1×10^{-6} of the exact solution; root-mean-square Hamiltonian constraint 2×10^{-6} of its largest term. Sources held fixed: lapse 8 % low, density 4 % high, constraint 0.15 of its largest term.

C1, amplitude 10^{-4}, thirty-two cells, to $a = 100$. The Hamiltonian-constraint norm of the code at the start is 6×10^{-10}. Conformal factor against the dictionary: 23 % of the perturbation off with particles at cell centers, 0.7 % with particles at cell corners. Observer density contrast: within 0.5 % of linear theory up to $a_p = 30$, +0.75 % and −0.70 % at $a_p = 100$ (contrast 0.03); the two theories differ by 0.3 % and 0.8 % of the contrast there.

C3 test, thirty-two cells. Redshift within 9×10^{-7}; luminosity distance low by 1.2×10^{-4} at emission at $a = 1$; measured and expected error agree to 6×10^{-7}.

C2, first runs (relativistic lattice displaced 4.04 times the Newtonian one). Deposited density at the probes 0.5 % below target at amplitude 0.01 and 7 to 14 % below at 0.05. First crossing of the fiducial observers later in the relativistic run by 5, 9, 14, and 23 to 28 % in a_p for amplitudes 0.0025, 0.005, 0.01, 0.05 (Ref. [1], read from its figures: 7, 9, 12 % for the three lower amplitudes). Largest amplitude: relativistic run ends at $a = 18.6$, Newtonian dictionary at $a = 14.4$. Constraint violation relative to its largest term up to $a = 4$: 3×10^{-4}, 1×10^{-2}, 1×10^{-1} for amplitudes 0.0025, 0.01, 0.05.

C2, identical positions, sixty-four cells, linear cloud.

TABLE I. Targets and state. C0 to C6 are the milestones of the plan; T1 and T2 are the method developments of Sec. V.

Target	Content	State
Plan	Sources read; design of slicing, initial data, reference code, observables	done
C0	Periodic particles, cosmology setup, proper time per particle, homogeneous-universe test	done
C1	Constraint-solved initial data on the torus; linear growth	done
C2	Ref. [1]: observer density contrasts and crossing times, four amplitudes, both gravities	done at 64 cells, three cloud shapes
C3	Ref. [1]: luminosity distance against redshift	done at 64 cells
C4	Ref. [1]: constraint convergence, resolution and particle number, extreme case	cloud shapes, wavelength and resolution tests done; extreme case: lapse driver works; refined runs fail at the collapse
C5	Ref. [3]: all tests	done: homogeneous tests, linear mode, random field in both boxes (32 to 128 and 64 to 128 cells), conservation test; 256 cells in the smaller box left out
C6	Ref. [5]: all tests	done without radiation; the radiation-dependent curves are modelled only
T1	Constraint-compatible deposit	clouds of three orders; compensation of the cloud on the deposited density to fourth order; spline interpolation of the metric; conservation test on the mesh; limiter tested without gain
T2	Evolving the departure from a known solution	not started

Contrast at $a_p = 1$, relativistic against Newtonian: -0.33, -0.53, -0.94, $-5.8\,\%$ at the overdensity and $+0.06$, $+0.24$, $+0.60$, $+3.7\,\%$ at the underdensity for amplitudes 0.0025, 0.005, 0.01, 0.05. Densities at the underdensity agree within 0.5, 1.0, 2.2 % for the three lower amplitudes. Delay of the infall: 1.37, 2.60, 5.15, 26 % ($5.2\,\bar{\delta}$); in coordinate time the same to 0.1 % of the scale factor. By cloud shape at amplitude 0.01, inner and outer observers: 5.2 and 5.3 (linear), 4.6 and 5.3 (quadratic), 4.7 and 5.3 % (cubic). Infall of the inner observers, relativistic run, amplitude 0.01: $a_p = 63.1$, 58.9, 59.6 for the three shapes. Read from the figures of Ref. [1]: 6 to 7, 6, 10 to 11 % for the observers and 1.9, 5.8, 6.9, 31 % at a density contrast of ten.

Constraint violation, amplitude 0.01, mean over the mesh up to $a = 4$: $H_r = 2.0\times10^{-4}$, 6.0×10^{-6}, 9.9×10^{-6} and $M_r = 3.8\times10^{-2}$, 1.6×10^{-3}, 1.9×10^{-4} for the linear, quadratic, cubic cloud; after the collapse $H_r \approx 10^{-2}$ for all.

Resolution, amplitude 0.01, quadratic cloud, 64, 96, 128 cells: relativistic infall of the inner observers at $a_p =$ 58.88, 58.18, 57.92 and of the outer ones at 79.36, 79.00, 78.86; Newtonian 56.29, 55.70, 55.48 (inner) and 75.40, 75.04, 74.88 (outer); delay 4.61, 4.44, 4.39 % (inner) and 5.25, 5.27, 5.31 % (outer). Mean H_r at $a = 2$, 10, 40: 4.7×10^{-6}, 6.4×10^{-5}, 2.3×10^{-4} at 64 cells and 1.8×10^{-6}, 4.5×10^{-5}, 2.1×10^{-4} at 128.

Wavelength test, amplitude 0.01, quadratic cloud: delay 1.1, 4.6 to 5.3, 28 to 30 % at 2, 4, 8 initial Hubble radii; delay over the depth of the initial potential 2.4, 2.75 (2.85 at amplitude 0.05), 3.9.

C3, sixty-four cells, linear cloud. Deviation of the luminosity distance from the homogeneous relation at $z = 0.1$, 0.5, 1, 2, relativistic run, with the values read from Ref. [1] in brackets. Amplitude 0.01, overdensity ray emitted at $a = 1$: -0.0069, -0.0043, -0.0042, -0.0108 (-0.0077, -0.0051, -0.0048, -0.0111); emitted at $a = 28$: $+0.654$, $+0.025$ ($+0.605$, $+0.006$); underdensity ray emitted at $a = 28$: -0.176, -0.189 (-0.161, -0.173). Amplitude 0.0025, overdensity ray emitted at $a = 163$ at $z = 0.1$: $+0.320$ ($+0.309$). Relativistic against Newtonian for rays emitted at $a = 1$: at most 6×10^{-4}, 2×10^{-4}, 7×10^{-5} for amplitudes 0.01, 0.005, 0.0025; largest for later rays at amplitude 0.01: 5 % (overdensity ray, $z = 0.03$) and 8 % (underdensity ray, $z = 0.73$).

C5. Homogeneous universe to $a = 1001$ at 8, 16, 32 cells, exact data: error of χ at $t = 100\,\mathrm{Mpc}/h$: 3.9×10^{-9}, 2.4×10^{-10}, 1.6×10^{-11} (fourth order); at the end 1.3×10^{-7}, 8.2×10^{-9}, 1.1×10^{-9} with the constant damping coefficient and 3.9×10^{-8}, 2.6×10^{-9}, 6.2×10^{-10} with damping at a fixed rate per conformal time (Ref. [3]: 2.6×10^{-9}, 1.6×10^{-10}, 9.8×10^{-12}). With noise: stable; largest Hamiltonian constraint from 1.2×10^{-5} at $t = 100$ to 3×10^{-11} at the end at 8 cells. Density error at the end at 8, 16, 32 cells: 6.9×10^{-4}, 2.3×10^{-4}, 6.5×10^{-5} with the stated momentum noise, 2.4×10^{-6}, 1.2×10^{-6}, 6.4×10^{-7} without it (Ref. [3]: 8.9×10^{-7}, 1.9×10^{-7}, 6.7×10^{-8}); at $t = 10$ without it: 9.0×10^{-9}, 2.3×10^{-9}, 5.5×10^{-10} (7.7×10^{-9}, 2.2×10^{-9}, 4.4×10^{-10}). Linear mode at 32 and 64 cells: potential constant to 1.7 and 0.44 % (4.9

and 1.0 %); density within 2.3×10^{-2} and 6.0×10^{-3} of linear theory at the end (3.6×10^{-2}, 9.1×10^{-3}); largest relative Hamiltonian constraint 8×10^{-8} and 1.4×10^{-8}.

C5, random field, larger box ($704\,h^{-1}$Mpc, power above $220\,h^{-1}$Mpc, linear to the end; 8 particles per cell at 32 and 64 cells, 1 at 128; quadratic cloud). Mean of H_r at $z = 100$, 10, 1, 0.02. Courant step alone: 5.2×10^{-6}, 3.6×10^{-5}, 6.9×10^{-5}, 8.1×10^{-5} at 32 cells and 3.4×10^{-7}, 2.2×10^{-6}, 4.3×10^{-6} at 64 cells (ratio 15 to 16). With the step limited to one percent of the expansion time (1,243 steps instead of 1,056 at 32 cells): 7.1×10^{-7}, 6.6×10^{-7}, 1.5×10^{-6}, 4.9×10^{-6} (32) and 1.8×10^{-7}, 1.9×10^{-7}, 4.2×10^{-7}, 5.3×10^{-6} (64); with the compensation in addition 5.0×10^{-8}, 1.3×10^{-7}, 2.0×10^{-6}, 9.5×10^{-6} (32) and 2.1×10^{-8}, 8.5×10^{-8}, 6.4×10^{-7}, 1.0×10^{-5} (64). Ref. [3]: 3.1×10^{-7}, 5.1×10^{-7}, 2.6×10^{-6}, 3.5×10^{-6} (32) and 9.1×10^{-8}, 8.9×10^{-8}, 9.8×10^{-8}, 1.2×10^{-7} (64). Mean of M_r with the compensation: 1.7×10^{-3}, 2.0×10^{-3}, 2.3×10^{-3}, 7.1×10^{-3} (32) and 6.0×10^{-4}, 5.3×10^{-4}, 1.3×10^{-3}, 9.6×10^{-3} (64); without: 3.8×10^{-3}, 1.6×10^{-3}, 1.4×10^{-3}, 2.4×10^{-3} and 1.2×10^{-3}, 5.2×10^{-4}, 1.0×10^{-3}, 5.4×10^{-3} (Ref. [3]: 1.8×10^{-2}, 1.3×10^{-2}, 9.1×10^{-3}, 7.8×10^{-3} and 4.9×10^{-3}, 4.1×10^{-3}, 4.2×10^{-3}, 4.3×10^{-3}). Largest departure of the power from linear theory up to the cut-off at $z = 100$, 10, 1, 0: 11, 21, 27, 29 % (32 cells) and 2.8, 5.5, 7.4, 8.2 % (64) without the compensation; 0.26, 0.37, 0.98, 1.8 % and 0.05, 0.24, 0.99, 1.8 % with it (rms contrast 0.04 and 0.08 at $z = 1$ and 0). By cloud at 32 cells below half the cut-off at $z = 10$: 5.4, 8.0, 10.4 % short for the linear, quadratic, cubic cloud. With the pass applied after the division by the volume element: 1.8 %. At 128 cells without compensation: $H_r = 3.9 \times 10^{-8}$, 1.7×10^{-7}, 2.1×10^{-6}, 8.8×10^{-6}; power within 0.3 % to $z = 10$. An earlier series had the box size of the input rounded to six digits; the mesh volume was then 3.8×10^{-7} too small, which appeared as a uniform H_r of 2.7×10^{-7} at the start that grew to 2×10^{-5}.

C5, random field, smaller box ($256\,h^{-1}$Mpc, power above $40\,h^{-1}$Mpc; rms contrast 5.5×10^{-4} at the start, 0.26 at $z = 1$ and 0.63 at $z = 0$ in the run), 64 cells, 8 particles per cell, quadratic cloud unless stated; mean H_r and mean M_r at $z = 100$, 10, 1, 0.02. No compensation: 9.5×10^{-8}, 1.2×10^{-6}, 1.3×10^{-4}, 6.9×10^{-4} and 1.1×10^{-3}, 1.3×10^{-3}, 4.9×10^{-2}, 0.13. Compensated: 2.2×10^{-8}, 1.7×10^{-6}, 2.7×10^{-4}, 2.0×10^{-3} and 1.3×10^{-3}, 2.6×10^{-3}, 0.20, 0.47. Cubic cloud, compensated: 3.2×10^{-8}, 1.5×10^{-6}, 1.0×10^{-4}, 1.4×10^{-3} and 5.4×10^{-4}, 1.7×10^{-3}, 7.7×10^{-2}, 0.33. Constant damping coefficient, no compensation: $H_r = 1.2 \times 10^{-7}$, 1.3×10^{-6}, 1.8×10^{-4}, 1.1×10^{-3}. Twenty-seven particles per cell, no compensation: 8.1×10^{-8}, 2.7×10^{-6}, 9.6×10^{-5}, 5.1×10^{-4} and 4.8×10^{-4}, 4.2×10^{-3}, 3.4×10^{-2}, 0.11. Limited compensation: as compensated up to $z = 0.3$, $H_r = 1.7 \times 10^{-3}$ at the end. Ref. [3] at 64 cells: 2.5×10^{-7}, 2.0×10^{-7}, 9.6×10^{-7}, 5.3×10^{-6} and 7.6×10^{-3}, 7.2×10^{-3}, 1.6×10^{-2}, 6.7×10^{-2}. Largest departure of the power from linear theory up to the cut-off at $z = 100$ and 10: 9.0 and 18 % without compensation, 0.09 and 0.53 % with it, 0.28 and 0.66 % with the cubic cloud (Ref. [3] at 64 cells: 0.9 to 13 % and 1.5 to 31 % across the band).

Sampling error of a displaced lattice (no evolution; field with waves down to ten cells; rms error of the density over the mean; linear, quadratic, cubic cloud at 2 particles per cell and direction): 1.9×10^{-3}, 1.0×10^{-4}, 8.7×10^{-6} at a largest contrast of 0.2; 8.8×10^{-3}, 7.1×10^{-4}, 7.3×10^{-5} at 0.5; 2.4×10^{-2}, 3.1×10^{-3}, 4.8×10^{-4} at 2. With 3 per cell and direction: 2.0×10^{-3}, 1.5×10^{-4}, 5.0×10^{-6} at 0.2 and 1.3×10^{-2}, 1.2×10^{-3}, 1.3×10^{-4} at 2. With sub-particles interpolated between Lagrangian neighbors (2 per direction on the lattice of 2): 2.3×10^{-3}, 2.6×10^{-4}, 2.4×10^{-4} at 0.2.

Cubic cloud with the compensation and the step limit, larger box; mean H_r and mean M_r at $z = 100$, 10, 1, 0.02. At 32 cells: 6.4×10^{-8}, 1.5×10^{-7}, 1.9×10^{-6}, 7.6×10^{-6} and 2.1×10^{-4}, 7.8×10^{-4}, 1.4×10^{-3}, 4.3×10^{-3}. At 64 cells: 3.5×10^{-8}, 8.5×10^{-8}, 1.7×10^{-7}, 6.8×10^{-7} and 2.2×10^{-4}, 2.3×10^{-4}, 2.4×10^{-4}, 7.3×10^{-4}; power up to the cut-off within 0.20, 0.45, 1.2, 2.0 % at $z = 100$, 10, 1, 0. Constraint field at $z = 0$: at 32 cells 90 % of its power lies between 4 and 6 times the fundamental wave number (cut-off of the initial spectrum: 3.2), with rms 1.8×10^{-5} of $16\pi\langle E\rangle$ and a correlation of 0.79 with the Laplacian of the squared contrast; at 64 cells that part has rms 4.8×10^{-7}, and 94 % of the remaining 2.2×10^{-6} lies above 13 times the fundamental.

Production configuration (cubic cloud, compensation with $c = 1/6$ and $c_2 = 7/15$, spline interpolation, step limit, damping rate 0.5), larger box, 32, 64, 128 cells (8, 8, 1 particles per cell). Mean H_r at $z = 100$, 10, 1, 0.02: 2.5×10^{-8}, 1.2×10^{-7}, 1.9×10^{-6}, 7.8×10^{-6}; 2.3×10^{-8}, 8.5×10^{-8}, 1.7×10^{-7}, 8.5×10^{-7}; 1.0×10^{-8}, 6.1×10^{-8}, 1.3×10^{-7}, 8.3×10^{-7} (Ref. [3]: 3.1×10^{-7} to 3.5×10^{-6}; 9×10^{-8} to 1.2×10^{-7}; 2.4×10^{-8} to 5.1×10^{-8}). Mean M_r: 2.6×10^{-4}, 7.9×10^{-4}, 1.5×10^{-3}, 4.4×10^{-3}; 2.6×10^{-4}, 2.3×10^{-4}, 2.4×10^{-4}, 8.7×10^{-4}; 5.5×10^{-5}, 5.7×10^{-5}, 9.2×10^{-5}, 7.9×10^{-4} (Ref. [3]: 1.8×10^{-2} to 7.8×10^{-3}; 4.9×10^{-3} to 4.3×10^{-3}; 1.2×10^{-3} to 2.2×10^{-3}). Power against linear theory, signed, for the three shells up to the cut-off, at 64 and at 128 cells: −0.006, +0.005, −0.012 and −0.006, +0.006, −0.009 % at $z = 100$; +0.067, +0.142, −0.015 and +0.070, +0.146, −0.007 % at $z = 10$; +0.47, +0.84, +0.00 and +0.48, +0.84, +0.01 % at $z = 1$; +0.96, +1.65, +0.06 and +0.97, +1.65, +0.07 % at $z = 0$ (rms contrast 0.008, 0.042, 0.084 at $z = 10$, 1, 0). Conservation on the mesh, momentum and energy residuals while the flow is smooth: 4×10^{-4} to 10^{-3} and 4.6×10^{-3} at 32 cells, 7×10^{-5} and 4×10^{-4} at 64, 5×10^{-6} and 3×10^{-5} to 2×10^{-4} at 128. Smaller box in this configuration, 64 and 128 cells (8 and 1 particles per cell). Mean H_r at $z = 100$, 30, 10, 5, 1, 0.02: 2.8×10^{-8}, 1.9×10^{-7}, 1.5×10^{-6}, 5.6×10^{-6}, 1.2×10^{-4}, 1.9×10^{-3} and 5.2×10^{-9}, 4.8×10^{-9}, 3.9×10^{-8}, 1.0×10^{-6}, 7.8×10^{-5}, 9.6×10^{-4} (128 cells without the new elements: 6.0×10^{-8}, 5.5×10^{-7}, 4.4×10^{-6}, 1.5×10^{-5}, 2.1×10^{-4}, 8.6×10^{-4}; Ref. [3] at 128 cells: 4.5×10^{-8} at $z = 100$, 4.3×10^{-8}

at 10, 1.2×10^{-7} at 5, 3.9×10^{-7} at 1, 4.8×10^{-6} at 0). Mean M_r at 128 cells: 1.5×10^{-4}, 1.6×10^{-4}, 2.2×10^{-4}, 1.4×10^{-3}, 6.3×10^{-2}, 0.30 (Ref. [3]: 2.0×10^{-3}, 2.8×10^{-3} at 10, 7.1×10^{-3} at 1, 4.2×10^{-2} at 0). Power against linear theory, six shells up to the cut-off, at 128 cells with the 64-cell values in brackets: $z = 100$: within 0.04 % (0.06); $z = 10$: +0.25, +0.43, −0.31, −0.37, −0.30, +0.08 (+0.24, +0.43, −0.32, −0.40, −0.35, −0.04) %; $z = 1$: +1.31, +2.08, −1.81, −1.21, −1.69, +5.62 (+1.31, +2.07, −1.82, −1.24, −1.76, +5.46) %; $z = 0$: +2.35, +3.16, −4.04, −0.68, −3.19, +20.2 (+2.34, +3.15, −4.05, −0.72, −3.27, +20.0) %; rms contrast 0.05, 0.30, 1.06 (0.88) at $z = 10$, 1, 0.

Random field in the Newtonian reference, larger box, 32 cells, same numerical configuration; power against the linear theory of each code, three shells: Newtonian +0.08, +0.13, −0.10 % at $z = 10$, +0.49, +0.82, −0.13 % at $z = 1$, +0.98, +1.62, −0.10 % at $z = 0$; relativistic +0.06, +0.11, −0.13; +0.47, +0.79, −0.17; +0.95, +1.60, −0.14 %; relativistic minus Newtonian −0.014 to −0.046 % in every shell and at every epoch.

Collapse at one wavelength, amplitude 0.01, 64 cells, quadratic cloud: with the spline interpolation the relativistic infall of the four groups of observers moves by −0.03, +0.05, 0.00, 0.00 % in a_p. Production configuration in both calculations (cubic cloud, compensation to fourth order, no filter; spline and step limit in the relativistic run), 64 cells: relativistic infall at $a_p = 58.07$, 58.66, 61.66, 78.77, Newtonian at 55.53, 56.13, 58.92, 74.83; delay 4.58, 4.51, 4.65, 5.27 %. Mean H_r at $a = 2$, 5, 10, 20, 40, 100: 3.0×10^{-7}, 2.1×10^{-6}, 3.7×10^{-6}, 3.2×10^{-5}, 2.4×10^{-4}, 1.6×10^{-2} (quadratic cloud with the filter: 4.9×10^{-6}, 8.1×10^{-6}, 6.9×10^{-5}, 1.1×10^{-4}, 2.3×10^{-4}, 4.0×10^{-3}).

Metric interpolation and conservation on the mesh. Gradient response of the Hermite reconstruction for a wave of ten cells: −6.4 % at the cell centers, +0.6 % at the quarter points (with a part of 1.3 % out of phase), +3.1 % midway between centers. One wavelength of 32 cells, linear cloud, one particle per cell at the cell corners: momentum residual 1.27×10^{-3} in every cell, predicted $[(1 - (k\Delta x)^2/8)(1 + (k\Delta x)^2/12) - 1]/\sqrt{2} = 1.15 \times 10^{-3}$; with the spline 3.5×10^{-3} without compensation, 1.3×10^{-4} with $c = 1/8$ (the response of the linear cloud deposited from the cell corners), and with the cubic cloud and $c = 1/6$: momentum residual 1.4×10^{-4}, energy residual 1.4×10^{-5} (4.2×10^{-3} with the linear cloud). Random field, larger box, cubic cloud: momentum residual 2.2×10^{-3} (Hermite) and 4.3×10^{-4} (spline) at 32 cells, 2.6×10^{-5} to 1.2×10^{-4} (spline) at 64 cells; energy residual 4.6×10^{-3} at 32 cells and 4×10^{-4} to 3.9×10^{-3} at 64 cells (Ref. [3]: 10^{-2} for both at all resolutions). Power against linear theory at 32 cells, signed, for the three shells up to the cut-off at $z = 100$ and 10 (the last shell is the wave of ten cells): Hermite with the three-point pass +0.08, +0.23, +0.19 and +0.35, +0.76, +0.53 %; spline with the three-point pass −0.06, −0.23, −0.62 and −0.01, −0.29, −1.26 %; spline with the fourth-order pass −0.01, −0.01, −0.06 and +0.06, +0.11, −0.13 %.

Damping, 32 cells, cubic cloud, compensated: mean H_r at $z = 100$, 10, 1, 0.02 is 2.0×10^{-7}, 4.9×10^{-7}, 6.3×10^{-6}, 2.2×10^{-5} at a rate of 2 per unit of conformal time; 6.4×10^{-8}, 1.5×10^{-7}, 1.9×10^{-6}, 7.6×10^{-6} at 0.5; 4.9×10^{-8}, 1.0×10^{-7}, 1.5×10^{-6}, 6.3×10^{-6} without damping. Unchanged at 32 cells by 27 particles per cell, by a tighter pusher tolerance, by half the step, and by a fifth of the dissipation. Fourth-order term of the compensation with the Hermite reconstruction: $H_r = 4.3 \times 10^{-8}$ (quadratic) and 7.2×10^{-8} (cubic) at $z = 100$; power of the shortest wave at $z = 10$ off by −0.37, +0.40, +0.53, +1.7 % for the quadratic and cubic clouds with the three-point pass and with the fourth-order term.

Initial data, 32 cells, larger box: solved conformal factor against the dictionary, longest wave: −5.4 % of the perturbation without compensation, −3.0 % with the pass after the division, −0.05 % with the pass on the density.

Step limit, homogeneous universe, 16 cells, box of four Hubble radii, to $a = 36$: $H_r = 2.5 \times 10^{-5}$ without (160 steps) and 6.0×10^{-8} with a limit of one percent of the expansion time (360 steps).

C6. Excess velocity power of the finite-difference velocity at offsets 1, 0.5, 0.25: 2.0003×10^{-4}, 5.0002×10^{-5}, 1.2500×10^{-5} at $z = 99$ and 8.0044×10^{-4}, 2.0002×10^{-4}, 4.9999×10^{-5} at $z = 49$ (Ref. [5]: 2.0×10^{-4}, 5.0×10^{-5}, 1.1×10^{-5} and 8.0×10^{-4}, 2.0×10^{-4}, 5.1×10^{-5}). Growth rate of the particle-number contrast one half at $k = 3.58 \times 10^{-3}$ and $5.06 \times 10^{-3}\, h$/Mpc (3.6×10^{-3}, 5.1×10^{-3}). Power ratio in the first bin of the large box: 52.9 and 16.9 (46.7 and 13.7; spread over realizations 38 to 57 and 13 to 18); small box 1.0553 and 1.0275 (1.049, 1.024). Synchronous growth rate on small scales: 1.0000 here, 0.9805 and 0.9903 with radiation in Ref. [5] (Mészáros estimate 0.9807, 0.9903). Scatter at the offset 0.05: 1.3×10^{-4} from spectra tabulated to seven digits, 1.6×10^{-7} from single-precision displacements (2 to 5×10^{-4}).

Extreme case with the lapse driver (strength one), sixty-four cells: lapse at the overdensity down to 7.5 % of the largest lapse at $a = 20.5$; the run ends at $a = 20.6$ (without the driver: 18.6) when the metric determinant turns non-positive in 48 cells around the overdensity.

Appendix B: Stated differences from the source papers

The formulation is Z4c with the lapse condition above and zero shift; Ref. [1] uses the generalized harmonic formulation in a damped harmonic gauge. Comparisons therefore use observer quantities only. The initial data hold the particles fixed in the constraint solve, where Ref. [2] holds the density fixed; the total rest mass is that of the dictionary, and the density has the dictionary shape up to a constant factor that differs from one at second order in the amplitude. The Newtonian reference

is a particle-mesh code at the resolution of the relativistic run, where Ref. [1] uses a tree code at higher force resolution. Particle numbers are lower than in the source papers, and the dependence on particle number is part of the tests.

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